CT PERFORMANCE EVAUATION REPORT - POST TEST OF ONE CELL (CT02) AFTER CHANGE OF DISTRIBUTION SYSTEM AND FAN

CT – 02 , CELL 02

OWNER – DHARIWAL INFRASTRUCTURE LTD CHANDRAPUR

TESTING AGENCY AND EQUIPEMNT PROVIDER - SS COOLING TOWER CONSULTANTS

DATE REPORT 15 APRIL 2026

DATE OF TEST 27 March 2026

DHARIWAL INFRASTRUCTURE LTD

CHANDRAPUR - 2 x 300 mw

CT - 2

REPORT – 15 APRIL 2026

PREAMBLE

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Due to poor performance of the cooling tower, factors influencing the poor performance were identified to be progressively resolved. To carry out this procedure it was decided to carry out the changes on one cell and based on the result carry out the same on balance cell. Accordingly one cell was isolated and water from the cell was collected at both ends and the temperature of the collected water was measured. This procedure had to be conducted both before change of gear box and fan and post change of gear box and fan (running at higher power) to ascertain the effect on cold water temperature on one single cell so that the same could be replicated in other cells.

Accordingly on 23 of September 2025, Pre upgrade test was carried out. testing team reached site and carried out collection of data from 4 to 8 pm . (stabilized peak load was available at 4 .0 pm and hence stable readings from 5 to 9 pm was collected. Most stable 1 hour readings are taken up for assessment / benchmarking to be compared with post gearbox and fan replacement measurements.

One page calculations are made available attached to this report

The existing Maya fan was replaced with another fan and new gear box of higher rating. The fan was operated at 116 kw. Based on these conditions a single cell Post change test was carried out on cell no 2 in exactly the similar manner to the pre test so that the improvement in the cell can be calculated. The test was carried out **on 6th October 2025 from 5 to 9 pm**

Report for the same has been submitted

Subsequently distribution system modifications were made and test was conducted on 27 March 2026 in a single isolated cell to understand effect of these changes.

ISOLATED CELL FOR WATER COLLECITION

MEMBERS PRESENT FOR THE TEST

MS DHARIWAL INFRASTRUCTURE

Mr Shrikant Shrivastava (Remote)

Mr GAURAV HANTODKAR

Mr PAWAN GAWANDE

MS SSCTC

Mr SURESH SARMA

Mr SANJAY GORAD

Mr MRADUL VISHWAKARMA

Mr PARAG VISHWAKARMA

Mr RAHUL MANKE

Assessment method and analysis

Pre test was conducted on 27 March 2026.

Cell no 2 has been isolated at the basin level for collection of cold water directly from the rain zone by providing sheets of frp placed on structural members at the top of the basin so as to collect water from the rain zone and move the water by slope towards the drums placed on either air inlet.

All required data was collected and report has been given.

Similarly on 6th October 2025 Post fan and gear box change test was conducted exactly in line with the pre test methods

24 nos RTD were placed on the drums for measuring cold water temperature

2 nos RTD were placed to measure hot water temperature

2 nos wet bulb station were placed , one near each air inlet to measure WBT and DBT.

Power was measured from the clients MCC

Air flow was measured for the fan on cell 2

Water flow was measured at the riser

Analysis Method

Since 2 different set of conditions have to be evaluated, there has to be a reference to which both these test collected data can be compared to.

This base we have taken as the design conditions

So pre test has been compared to design conditions and post test has been compared to design conditions.

Accordingly the difference between pre and post test has been established.

Suresh Sarma
sarmasuresh1960@gmail.com
+91 99875 49712

Instrument placement and readings taken

Air flow was measured using Data Logging anemometer, manually generally as per CTI ATC 143 Method of equal area.. Totally 10 traverse were taken in each quadrant thus total of 40 readings were taken for the one fan.

Hot water – was taken at inlet of the hot water to the cooling tower which is common for all cells.

Cold water – As water was collected using a sheets supported on structure and leading the water to drums placed at either air inlet side. 12 sensors were placed on side A and 12 sensors were placed on side B. thus 24 sensors were used to measure the temperature of the water collected.

Water flow – was measured using UFM (GE Make) on the riser.

DBT – was measured and noted for reference

WBT (inlet) -was measured on either side of the air inlet using wet bulb automatic stations recording wbt every minute.

Power – to the fan motor was noted from the MCC of client

Suresh Sarma
sarmasuresh1960@gmail.com
+91 99875 49712

CONCLUSION AND REMARKS

1. Ms Dhariwal Infrastructure Ltd – Chandrapur, had been experiencing shortfall in cooling tower performance.
2. With a view to identify shortfall and prepare and action plan to enable Thermal performance improvement of the cooling tower. Testing on the basis of CTI ATC 105, was carried out.
3. Cooling Tower Performance test conducted on cooling tower CT2 in March 2025 showed 5.7 deg C cwt deviation – as per procedure – CTI ATC 105 By SSCTC
4. Probable reasons for the same and modifications for improvement were suggested by SSCTC
5. With a view to projecting and assessing a single cell performance which could be replicated to multi cells, bench marking of single cell cooling tower was carried out, which was referred to as Pre Test on September 2025
6. Subsequently since Fan performance was identified as one reason for performance shortfall, the fan was changed to original fan and a test was conducted which was referred to as Post fan replacement single cell test. Large improvement was observed to about 1.5 deg C which was toned down to about 0.7 deg C considering various uncertainties in the system.
7. A second modification was carried out in the form of re positioning the distribution system for better uniform water spray to the fills. (2 layers of fills were removed for this purpose)
8. Testing of the cell was carried out after incorporating the distribution system changes on 27th of March 2026. The additional head available in the system caused flow to increase substantially in the cell. Even under these conditions the cell showed an improvement in performance of approx. 0.5 deg C above the post fan modification test and 1.9 deg C above the initial as is existing singe cell pre test carried out.
9. All testing was carried out under 3 pump working conditions
10. Testing of one cell after isolation has higher uncertainties as compared to full cooling tower testing. However individual cell testing is less expensive and can be a very good indicator for the measured performance in comparison to existing other cell performance. This method thus is a very economical method for understanding cell performance after component changes leading to structured way of cooling tower performance improvement process with minimum of cost. Especially in the present layout of cooling tower at Chandrapur plant where only one outlet is available for the complete cooling tower. Generally in

Suresh Sarma
sarmasuresh1960@gmail.com
+91 99875 49712

multicell arrangement of cells the basin is divided into 2 halves (in more than 90%) of the cases and under such conditions half of the cells can be checked for improvement. However in present case individual cell isolation and assessment was assessed as the most practical and economic method.

11. The uncertainties arise from 1. The number of streamlets being tested. 2 the high turbulence at the point of measurement.3. instantaneous temperature changes of the streamlets.4) variations of water flow across the plan area of the tower leading to loss in performance.
12. Considering uncertainties exist in testing based on one cell. The expected improvement after changing of the distribution system for the cooling tower as predicted by us would be 1.1 deg C
13. Note – 60 Nozzles are still old Paharpur type nozzle, and would be causing loss in performance as it would not be in conjunction with the spray provided by other nozzles. Suggest change all as soon as possible. The effect of these nozzles during the pre test stage was very limited as the head was much lower and spray was hitting the structural beams thus restricting the spread from the nozzle, but post distribution the effect would be much higher.
14. Based on lab teste results, V bar offers high resistance to air flow specifically at higher water loadings. This could also be due to cascading effect of water due to specific shape of V bar placed in counterflow towers. Hence the performance of the tower is likely to be further increased than calculated values due to lower water loading of V bar after changes of distribution components in more number of cells.
15. Fan air flow has naturally decreased for the post distribution change as the water flow has increased by about 23%. New fan air flow values are provided within this document. i.e. 453 m3/s – Measured by Client Anemometer and 467 m3/s – Measured by SSCTC Anemometer.
16. For actual values of various tests carried out individual reports have been submitted which may be referred to as reference.

Suggestions and special notes

17. With 3 pumps operating and one cell under higher flow velocities through the duct would be higher by 1.23% this would naturally cause a higher static pressure at the back end of the cooling tower away from the riser. To mitigate this one sizer lower ferrul (orifice) can be provided at the last 3 pipes of the cell away from the riser. This would help provide uniform flow across the tower at higher heads. Ferruls are small frustrum of cone type projections put inside nozzle . the cost of a ferrul is about 10 – 15 Rs .
18. However as the number of cells with changed distribution increases to more than 5 this system would need to be re assessed and maybe ferruls may need to be removed as the flow quantity reduces in each cell due to

Suresh Sarma
sarmasuresh1960@gmail.com
+91 99875 49712

head equalization.

19. As already discussed during our many interactions the Nozzle provided by Brentwood is a standard Hamon Make French Sprayer freely available in the market.
20. Details in form of table is provided in next page for all the 3 tests carried out.
21. A general comparison has been made based on the cold water temperature measured at the tested cell and at the common cw channel. It is a general thumb rule that about 0.8 deg C is increase in cold water temperature for 10% increase in water flow to the cooling tower cell. Hence since the average flow through the test cell is 23% higher than all the other cells the temperature of the tested cell on identical conditions should have been roughly 2 deg C higher than the cw channel . however observations show 0.6 to 0.7 degree higher colder water at the tested cell indicating a 1.2 to 1.3 degree better performance for the cell even at higher water loading causing poorer performance of V bar.

Suggestion for further improvement –

1. Providing a vertical barrier near the water collection location to reduce the effect of water velocity and thus turbulence at point of measurement.
2. Placing one single large pipe to have higher accurate temperature distribution and measurement.

FINAL DATA AND CONCLUSION OF PRE AND POST TEST						
	TEST 1			TEST 2		TEST 3-1920hrs
		PRETEST		POST TEST		POST TEST
				Fan Change		Distribution change
Water Flow	M3/hr	2998		3067.21		3680
WBT	Deg.C	25.25		24.22		21.7
DBT	Deg.C	27.52		24.38		33.51
HWT	Deg.C	44.67		43.21		42.13
CWT	Deg.C	35.08		32.89		32.4
Fan Power At Motor Inlet	KW	97.04		116.24		117
Fan Air Flow	M3/s	405.97		499		485
Range	Deg.C	9.59		10.32		9.73
Approach	Deg.C	9.83		8.67		10.7
CWT Devaition from Design	Deg.C	5.8		4.3		3.83
Improvement	Deg.C			1.5		0.47
Improvement "3" wrt "2"				0.7		0.42
Improvement "3" wrt "1"	Deg.C					1.97

Note:

1.Improvement of 1.5 appears high in "TEST 2" so we consider that as 0.7 Deg C
times the original water loading
distortion and cascade effect
4.single cell isolated testing involves "UNCERAINITY" in measurement due to turbulent water flow
5.during the " TEST 3" Water flow was 23% higher, for splash bars the performance at Highwer water decreases rapidly.
6.It is expected that when more cells are provided with Distribution change the flow will reduce and performance will further improve.
7.Hence taking tolerance of 0.6 deg C as "UNCERTAINITY" we can predict that Min improvement should be
8.Actual improvement will definitely be greater than the above value

9.Theoretical total improvemen is	1.97	Deg C
10.Practical Improvement (without uncerainity)	1.17	Deg C
11.CWT channel temp (1940 to 2040)	31.76	Deg C
12.So even at 23% higher the teste cell had CWT	32.4	Deg C
13.so at 23% the cwt should have been	33.76	Deg C
14.Improvement (based on common channel)	1.36	Deg C

SSCTC forecast

15. Reference - As per original test single cell carried out , at 25.5 deg C original test cwt = 35.1Deg C
15. Reference - After all cells refurbishment carried out, at 25.5 deg C cwt predicted = 32.8Deg C

note : If further improvement is required - change fills to Golden Grid splash fill

Expected Improvement For CT after changes to all cells - Deg C 1.1

S S
COOLING TOWER
CONSULTANTS

Suresh Sarma
sarmasuresh1960@gmail.com
+91 99875 49712

Thermal Design, CT Assessment and upgrade, CT Testing, Technical Management & Training
7 Uma Sada- RCHS, CA Road, Chembur, Mumbai – 400071. www.ssctc.org

Air flow using Client Anemometer

			DIRECTION 1				DIRECTION 2				DIRECTION 3				DIRECTION 4		
Sr.No	DISTANCE FROM FAN RING		V1	v2	VAVG		V1	v2	VAVG		V1	v2	VAVG		V1	v2	VAVG
	m	ft	HIGH	LOW	M/S		HIGH	LOW	M/S		HIGH	LOW	M/S		HIGH	LOW	M/S
			M/S	M/S			M/S	M/S			M/S	M/S			M/S	M/S	
1	0.14	0.45	5.76	3.66	4.71		4.73	2.98	3.86		6.49	3.70	5.10		6.85	4.61	5.73
2	0.42	1.39	6.15	4.25	5.20		5.43	3.26	4.35		8.29	4.03	6.16		6.35	3.97	5.16
3	0.73	2.39	7.48	4.66	6.07		6.99	3.14	5.07		8.11	4.39	6.25		8.02	6.12	7.07
4	1.05	3.45	9.42	5.54	7.48		9.01	6.58	7.80		9.77	6.33	8.05		8.35	5.29	6.82
5	1.40	4.60	10.55	7.11	8.83		9.54	6.85	8.20		10.03	6.82	8.43		10.73	7.62	9.18
6	1.78	5.86	11.49	6.60	9.05		10.19	7.17	8.68		10.57	4.95	7.76		10.91	6.47	8.69
7	2.21	7.25	8.82	4.56	6.69		8.02	5.48	6.75		9.10	1.46	5.28		10.32	5.61	7.97
8	2.70	8.86	3.08	0.00	-1.08		8.41	1.87	5.14		5.10	-2.23	1.44		7.86	-0.73	3.57
9	3.29	####	0.00	0.00	0.00		0.00	0.00	0.00		0.00	0.00	0.00		0.00	0.00	0.00
10	4.11	####	0.00	0.00	0.00		0.00	0.00	0.00		0.00	0.00	0.00		0.00	0.00	0.00
	AVERAGE VELOCITY				4.69				4.98				4.85				5.42

1	Average Velocity(v)	4.99	m/s	3	Corrected Velocity(v1)	4.99	m/s
2	Calibration Constant K	1		4	Total Free Area	92.25	m2
				5	Total Flow = 3 X 5	459.84	m3/s

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Air flow using SSCTC Anemometer

ssctc anemometer

Stack Diamete	m		11.1		m	HUB DIA		2.4		M	CELL NO						
Fan Diameter	m		10.36		m	DATE OF TEST		27-Mar-26			UNIT						
Fan Make			HAMON			MOTOR RATING		132		KW							
Place of Testing			Dhariwal Chandrapur			MOTOR POWER		117		KW							
			DIRECTION 1			DIRECTION 2				DIRECTION 3				DIRECTION 4			
Sr.No	DISTANCE FROM FAN RING				VAVG			VAVG				VAVG				VAVG	
	m	ft			M/S			M/S				M/S				M/S	
1	0.14	0.45			3.45			4.63				3.73				5.65	
2	0.42	1.39			3.80			4.03				3.93				5.94	
3	0.73	2.39			4.95			5.37				4.83				6.29	
4	1.05	3.45			5.18			6.68				6.29				6.32	
5	1.40	4.60			6.97			7.73				7.24				6.06	
6	1.78	5.86			8.04			7.93				7.53				4.62	
7	2.21	7.25			7.58			6.79				7.37				2.81	
8	2.70	8.86			3.69			3.55				3.65				2.43	
9	3.29	10.80			0.00			0.00				0.00				0.00	
10	4.11	13.48			0.00			0.00				0.00				0.00	
	AVERAGE VELOCITY				**4.36**			**4.67**				**4.46**				**4.01**	

1	Average Velocity(v)	**4.38**	m/s	3	Corrected Velocity(v1)	5.06	m/s
2	Calibration Constant K =	1.16		4	Total Free Area	92.25	m2
				5	Total Flow = 3 X 5	467.02	m3/s

S.S COOLING TOWER CONSULTANTS
Suresh Sarma
sarmasuresh1960@gmail.com
+91 99875 49712

DHARIWAL CHANDRAPUR 2 X 300 MW - TPP - ONE CELL TEST AFTER DISTRIBUTION CHANGE

Test Date - 27-03-2026

COOLING TOWER TESTING BY SSCTC

1920 - 2020 HRS

15-Apr-26

1940

		Design Values	Recorded Tes readings	
Water Flow Rate	M3/hr	3863.6		3680.0
Hot Water Temp.	Deg C	43		42.13
Cold Water Temp	Deg C	33		32.4
Wet Bulb Temp	Deg C	29		21.7
Fan Driver Power	KW	117		117
Range	Deg C	10		9.73
Approach	Deg C	4		10.7

STEP 1

3 sheets of performance curves are provided for 90%, 100% and 110% water flows.

In each of the curve draw a vertical line from design wbt to intersect all 3 Ranges as shown below.

Prepare a table of "COLD WATER" vs "RANGE" for each of the 3 flows at test WBT.

(Performance curves are attached along with the submission of report)

FIGURE 1

FIG- 1 Curve at 100% flow intersecting with 21.7 Deg C "WBT" with all 3 Range Values

In above curve the intersection of wbt with 80%,100 & 110% range are at 27.81 29.02 30.11

(All 3 curves 90%,100% and 110% flow with the intersection points are attached with this report.)

Cold Water "Temp"vs"Range" for Test Wet Bulb Temp of 21.7 Deg.C
ranges.

Similarly plot for 90% water flow and 110 % water flow

find out the cold water at all intertsections and make a Table - 1 as below

TABLE (READINGS AT TEST WBT OF) 21.7

FLOW →	3477	3864	4250
Range	CWT(Deg.C)	CWT(Deg.C)	CWT(Deg.C)

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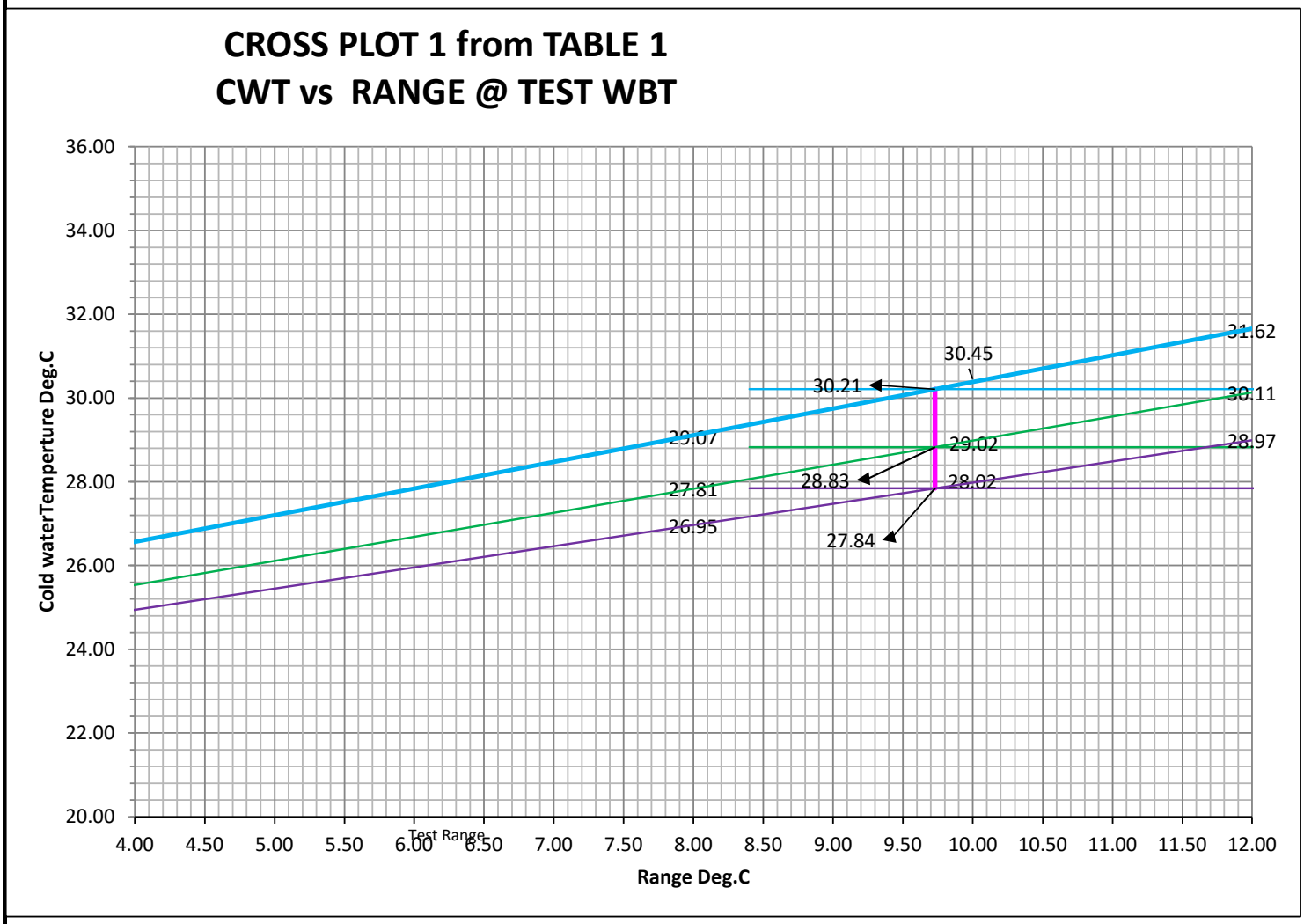
Suresh Sarma
sarmasuresh1960@gmail.com
+91 99875 49712

8.00	26.95	27.81	29.07
10.00	28.02	29.02	30.45
12.00	28.97	30.11	31.62
Table of cold water temp vs range for 3 w	90%Flow	100%flow	110%flow

STEP 2

Plot CROSS PLOT 1 from "TABLE 1" with "CWT" in Y axis and "TEST RANGE" on X axis



REFER To Cross Plot 1 above " Range vs Cold Water Temperature. Design Range plotted"

From cross plot 1 mark the test range in the curve above to intersect each water flow at 3 cwt points (draw red line)

From cross plot above generate Table 2 as below **(intersection of purple,Green&blue curve with red range vertical line)**

REFER TO CROSS PLOT 1 ABOVE **TABLE 2**

Cold Water Temp Vs. Water Flow @range	9.73	Deg.C		
Water Flow X 1000 (M3/HR)	3.477272727	3.8636364	4.25	4.636364
Cold water Temp	**27.84**	**28.83**	**30.21**	
	PURPLE LINE	**GREEN LINE**	**BLUE LINE**	

INTERSECTION POINT OF RANGE (PINK) LINE

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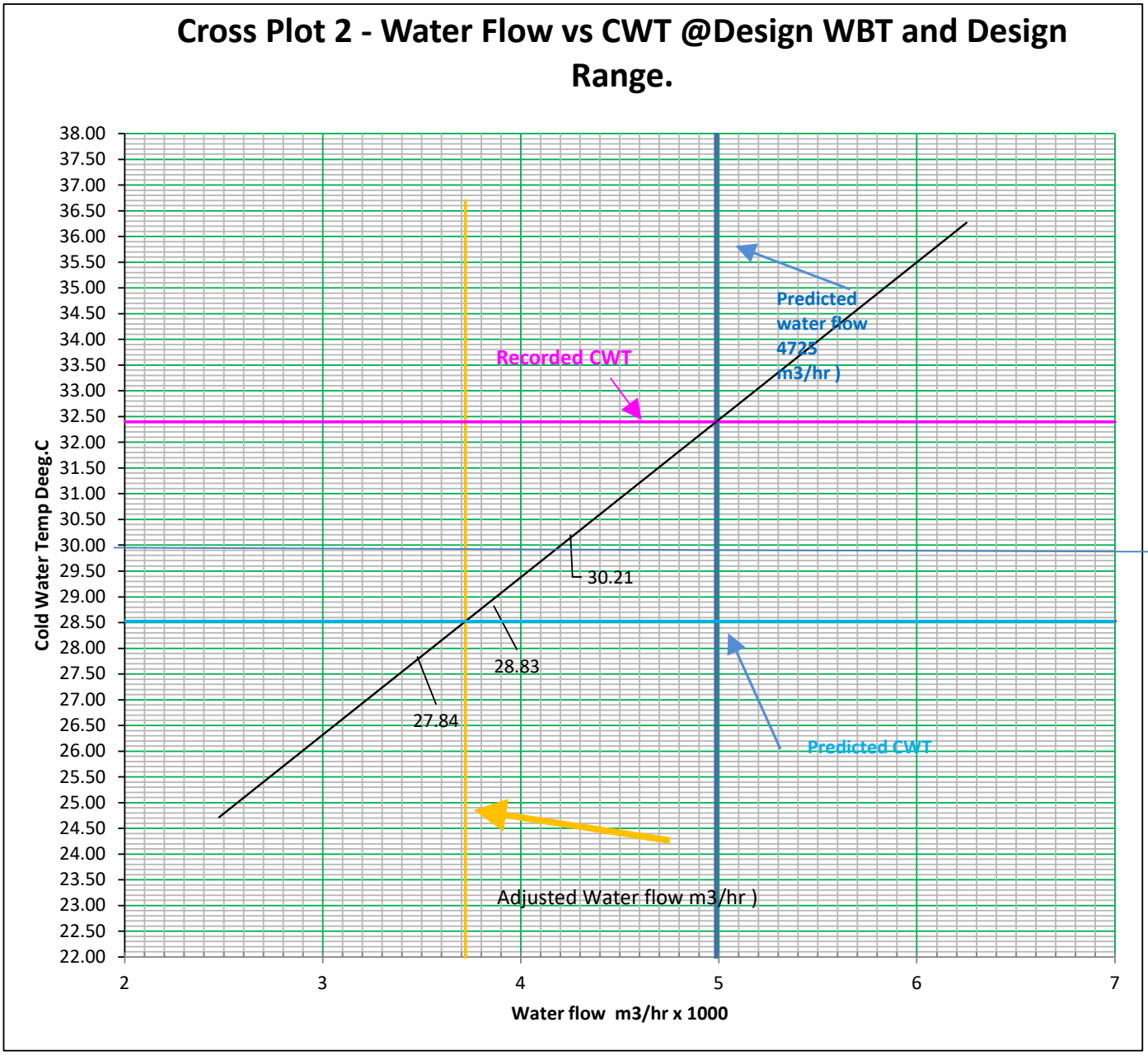
Suresh Sarma
sarmasuresh1960@gmail.com
+91 99875 49712

STEP 3 : Now based on Table 2 plot a graph of cold water on Y axis with water flow on X axis



STEP 4

Calculation for Adjusted Water Flow:

a	Fan Power Design (Wd)	117	KW
b	Fan Power Test (Wt)	117	KW
c	Hot Water Temp.(Test)	42.13	Deg C
d	Cold Water Temp.(Test)	32.4	Deg C
e	Range(Test)	9.73	Deg K
f	WBT(Test)	21.7	Deg C
g	Approach(Test)	10.7	Deg C
h	Test Flow (Qwt)	3680	M3/hr
i	Density Test / Density Design	1.0337	

Adjusted Flow (Qwt(adj)) Qwtest x $(Wd/Wt)^{(1/3)}$ X (Den(t)/(Den(d))^1/3

Adjusted Flow (Qwt(adj)) h x (b/c)^(1/3)*i^(1/3)

= 3720.91 m3/hr

S.S COOLING TOWER CONSULTANTS
Suresh Sarma
sarmasuresh1960@gmail.com
+91 99875 49712

Step 5

i	Now in the cross plot "2" the curve extends up to 110% of Design water flow =	4250	m3/hr		
ii	Mark the Adjusted water flow in X axis and project the line to intersect the graph of FLOW vs CWT				
	the adjusted water flow in the x axis o	3720.91	m3/hr		
iii	From the point of adusted water flow draw a vertical line to intersect the cross plot , read the CWT horizontally				
iv	the value of cold water as read from the above is	**28.57**	Deg.C		
v	Cold Water Deviation =	Predicted Cold Water (expected)	-	Test(recorded) Cold Water	
vi	From cross plot 2 the Predicted(expected) Cold water =	**28.62**	Deg.C	(Refer cross plot 2 above)	
vii	**Tolerance on CWT**	0.00	Deg.C	(SPECIFICATION)	
viii	al Predicted Cold Water Temperature =(a	28.62	Deg.C	(pt a + pt b)	Tolerance of 0.3 Deg C not considered
ix	Test Cold Water (Recorded site data)	32.40	Deg.C	(Recorded)	
x	Predicted Water flow	4976.65	m3/hr		
x1	From above the "Test CWT" is higher than "Predicted CWT" by	3.83	Deg.C		

CONCLUSION

Capability (%) = (adjusted water flow / predicted water flow)x100 74.8 %

In above case the "Test Cold Water " is higher than Predicted Cold Water by **3.83** **DegC**

Hence cooling tower shortfall in performance is **3.83** **DegC**